



Can We Create Electronic Plants?

The manufacturing of analogue and digital organic electronic circuits and devices in living plants has been reported. The four key components of a circuit (an energy source, a conductor (wire), an electrical load, and at least one controller) have been achieved



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Plants in their various forms are always an integral part of the human life system, supporting it in different ways. Plants are the primary source of food and providers of oxygen. They sequester carbon and regulate the ecosystem; technologically, they synthesise materials and convert sunlight to chemical energy. They have non-speaking human-like complex organs that rely on the transport of ionic signals and hormones to perform necessary functions for them. However, plants operate on a much lower capacity and

slower time scale, thus making interacting with and studying plants difficult.

Break the barriers between the natural and the artificial

Scientists are developing bio-electronic technologies that will enable them to better understand how plants function and their biological processes. By functionalising plants with electronic materials, scientists and technologists aim to push the boundaries of knowledge and understand better the interface between the natural and the artificial.